

ZHUYUAN LIU

Spatial Capture ·
Digital Twin · Real-Time Media

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Full right to work in UK:
No sponsorship required

SKILLS:

SPATIAL CAPTURE & RECONSTRUCTION

LiDAR
Photogrammetry
Gaussian Splatting
3D Scanning
Digital Twins
Point Cloud Processing
Spatial Reconstruction
Blender
Unreal Engine

COMPUTATION & REAL-TIME

Python
p5.js
Computer Vision
Interactive Systems
Motion Capture
Object Tracking
Generative Media
AI-Assisted Workflows

DESIGN & PROTOTYPING

Rhino
Revit
Grasshopper
Adobe Creative Suite
Physical Computing
Arduino / ESP32
Rapid Prototyping
Digital Fabrication

SELECTED EXPERIENCE:

Sacred Intelligence — *Cultural Heritage Capture & Spatial Systems* 07/2026

Architectural Association Diploma

Conducted field-based 3D capture across Tibetan monastery environments using Gaussian Splatting as part of a wider investigation into cultural heritage documentation and digital preservation. The project combined spatial capture with architectural research, Unreal Engine environments, robotics, responsive media and physical-digital prototyping.

Within Katla — *LiDAR & Gaussian Splatting Landscape Capture* 03/2026

Collaboration with artist Camille

Contributed to the digital capture of an Icelandic landscape using LiDAR and Gaussian Splatting for Camille's project Within Katla, supporting the creation of a spatial archive and testing workflows for landscape-scale documentation and reconstruction.

No Curtain Call — *Digital Twin & Immersive Theatre* 07/2025

Architectural Association Diploma

Year-long collaborative project developed with theatre director Isabel and performer Milo, combining physical performance, spatial design and a real-time digital twin. Developed the scanned environment through photogrammetry-based reconstruction, followed by manual model refinement and rigging in Blender. Integrated inertial motion capture and AprilTag object tracking to connect performers and physical stage elements with the Unreal Engine environment in real time.

WORK EXPERIENCE:

Foster + Partners 07/2023 – 09/2024

Worked across competition and design-development projects within multidisciplinary teams, contributing to coordinated digital models, design iteration and real-time visual communication. Took responsibility for complex Revit workflows and translated coordinated spatial data into Unreal Engine environments for real-time visualisation and design communication.

Amateurist Studio 07/2022 – 03/2025

Co-founder. Led the design and delivery of a concert stage for YEANA and developed collaborative product and visual design projects.

EDUCATION:

Architectural Association School of Architecture 09/2024 – 06/2026

MArch Architecture, RIBA Part 2

Architectural Association School of Architecture 09/2020 – 06/2023

BA (Hons) Architecture, RIBA Part 1

Academy of Arts & Design, Tsinghua University 07/2014 – 07/2018

BA Sculpture

AWARDS & SCHOLARSHIPS:

Ars Electronica Film Selection 09/2025

Light Up New York for But a Moment, selected for **PANIC: Ars Electronica Animation Festival 2025**, and *Animation Festival on Tour 2026*.

AA Scholarship 08/2024

Awarded an Architectural Association scholarship for academic excellence in the Diploma Programme.